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DS PROGRAMMING

Kaviya R

192521075RD

QUESTIONS 10 / 10 completed

Structure

STRUCTURE

Write a C program to store student details (Roll No, Name, Marks) using structure and save them into a file. Display highest mark, class average, and total records stored.

Sample Input:
Enter number of

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 struct student{
3     int id;
4 };
5 int main(){
6     struct student s;
7     scanf("%d",&s.id);
8     printf("%d",s.id);
9     return 0;
10 }
```

INPUT

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QUESTIONS 10 / 10 completed

Array

ARRAY

Write a C program to find the 5th element in the given array/
Sample Input: a=[10,20,30,40,50,60]
Sample output: 50

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[6]={10,20,30,40,50,60};
5     printf("%d",a[4]);
6     return 0;
7 }
```

INPUT

Your INPUT goes here!

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QUESTIONS 10 / 10 completed

Fibonacci Series

FIBONACCI SERIES

Write a C program to print the first n terms of the Fibonacci series (n ≤ 20) without using recursion. Start with 0 and 1.
Sample Input/Output:
Enter number of terms: 8
Fibonacci series: 0 1 1 2 3 5 8 13

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,a=0,b=1,c,i;
5     printf("enter number of terms:");
6     scanf("%d",&n);
7     for(i=1;i<=n;i++)
8     {
9         printf("%d",a);
10        c=a+b;
11        a=b;
12        b=c;
13    }
14    return 0;
15 }
```

INPUT

8

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QUESTIONS 10 / 10 completed

Fib series-iterative ▾

FIB SERIES-ITERATIVE & RECUR

Write a C program to generate Fibonacci series up to N terms using both iterative and recursive methods. The program should display both outputs and compare execution time in milliseconds. Also display total recursive calls made.

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include<stdio.h>
2 int main(){
3     int n,a=0,b=1,c,i;
4     scanf("%d",&n);
5     for(i=0;i<n;i++){
6         printf("%d",a);
7         c=a+b;
8         a=b;
9         b=c;
10    }
11    return 0;
12 }
```

INPUT

5

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QUESTIONS 10 / 10 completed

Factorial

FACTORIAL

"Write a C program to find the factorial of a given number using recursion.
Sample Input: Enter a number- 6
Sample Output: Factorial of 6 is 720
"

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int fact(int n)
3 {
4     if(n==0)
5         return 1;
6     else
7         return n*fact(n-1);
8 }
9 int main()
10 {
11     int n;
12     printf("enter a number:");
13     scanf("%d",&n);
14     printf("factorial =%d",fact(n));
15     return 0;
16 }
```

INPUT

Your INPUT goes here!





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QUESTIONS 10 / 10 completed

Fibonacci series (N ▾)

FIBONACCI SERIES (N ≤ 50)

Develop a C program to print the first N terms of the Fibonacci series (N ≤ 50) without using recursion. Use two variables to generate the series iteratively and store the series in an array. The program should also calculate and display the sum of

PROGRAM EDITOR

C ▾

Run

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```
1 #include<stdio.h>
2 int main(){
3     int n,a=0,b=1,c,i;
4     scanf("%d",&n);
5     for (i=1;i<=n;i++){
6         printf("%d",a);
7         c=a+b;
8         a=b;
9         b=c;
10    }
11    return 0;
12 }
```

INPUT

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⌵

QUESTIONS 10 / 10 completed

Unsorted Array ⌵

UNSORTED ARRAY

Write a program to find the missing element for the given unsorted array of sequence numbers.
Sample Input: a=[10,9,7,4,3,1,2]
Sample output: 5,6,8

PROGRAM EDITOR

C ⌵

Run

Save

```
1 #include<stdio.h>
2 int main(){
3     int a[50],n,i;
4     scanf("%d",&n);
5     for(i=0;i<n;i++)
6         scanf("%d",&a[i]);
7     printf("unsotred array:");
8     for(i=0;i<n;i++)
9         printf("%d",a[i]);
10    return 0;
11 }
12
```

INPUT

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10 4 8 2 6

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QUESTIONS 9 / 10 completed

Linear Search ▾

LINEAR SEARCH

Create a C program to perform Linear Search on an array of 10 integers. The program should allow manual input or random generation of array elements. It must display the first position of the element (1-based indexing), total

PROGRAM EDITOR

C ▾

Run

Save

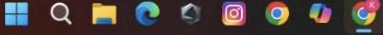
```
1 #include<stdio.h>
2 int main(){
3     int a[5],i,key;
4     for(i=0;i<5;i++)
5         scanf("%d",&a[i]);
6     scanf("%d",key);
7     for(i=0;i<5;i++){
8         if(a[i]==key){
9             printf("found");
10            return 0;
11        }
12    }
13    printf("not found");
14    return 0;
15 }
```

INPUT

4 5 26 3 8

3

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